Variational lattice-gap bounds: plaquettes, electric trials and holonomies

For a specified smooth gauge-invariant link function, the Feynman–Bijl identity bounds the physical Kogut–Susskind gap above by its electric gradient norm over its variance. Plaquettes give the strong-coupling order; an averaged spatial Polyakov loop gives a complementary exact susceptibility bound. Replacing its expectations by constant-mode values is a separate unproved small-volume step.

An abelian electric trial gives a structure-factor gaplessness criterion. For a **semisimple** gauge Lie algebra, a nonzero electric trial with constant colour coefficients is excluded by Gauss's law. This exclusion neither proves a positive gap nor a universal string energy for all dressed trials. Continuum smearing and positive-measure moments have their own canonical treatment.

The Feynman–Bijl inequality

Let H be self-adjoint and bounded below with simple ground state ψ_0 and next eigenvalue E_1, and let O be symmetric with $\psi_0 \in D(O)$, $O\psi_0 \in D(H)$ and $\langle \psi_0, O\psi_0\rangle = 0$. Then $O\psi_0 \perp \psi_0$, so by the min-max principle

$$E_1 - E_0 \;\le\; \frac{\langle O\psi_0, (H-E_0)O\psi_0\rangle}{\|O\psi_0\|^2} = \frac{\frac12\langle \psi_0, [O,[H,O]]\psi_0\rangle}{\langle \psi_0, O^2\psi_0\rangle},$$

using $[O,[H,O]] = 2OHO - O^2H - HO^2$ and $(H-E_0)\psi_0 = 0$. If O commutes with the Gauss generators, $O\psi_0$ lies in the physical sector and the same bound holds for Δ^{phys}. (Feynman, Phys. Rev. 94 (1954) 262, metadata; the argument is the two lines above.)

Link functions and strong-coupling order

Take $H = \frac{\hbar c}{a}[\frac{g^2}{2}\sum_\ell(-\Delta_\ell) + \frac{2}{g^2}\sum_p(N - \operatorname{Re}\operatorname{tr} U_p)]$ as in the obligations map §1 and a real function $O = f(U)$ with $\langle f\rangle = 0$. Then $[V, f] = 0$, and for the Laplacian $\Delta_\ell = \sum_a X_a^2$ in the left-invariant fields X_a of link ℓ,

$$[f,[-\Delta_\ell, f]] = 2\sum_a (X_a f)^2 =: 2|\nabla_\ell f|^2,$$

because $[\Delta, f]h = (\Delta f)h + 2\sum_a (X_a f)(X_a h)$ and the second commutator with f removes the (Δf) terms and leaves $-2\sum_a (X_a f)^2 h$. Hence

Proposition 1. For every real gauge-invariant $f(U)$ with $\langle f\rangle_0 = 0$,

$$\Delta^{\mathrm{phys}}_{a,L} \;\le\; \frac{\hbar c g^2}{2a}\,\frac{\sum_\ell \langle |\nabla_\ell f|^2\rangle_0}{\langle f^2\rangle_0}.$$

Corollary 2 (plaquette). With $f = \operatorname{Re}\operatorname{tr} U_p - \langle \operatorname{Re}\operatorname{tr} U_p\rangle_0$: for a link ℓ of p, $X_a \operatorname{tr} U_p = \operatorname{tr}(T^a M)$ up to a phase, with M a cyclic rearrangement of U_p, so by the $SU(N)$ completeness relation $\sum_a \operatorname{tr}(T^a M)\operatorname{tr}(T^a M^\dagger) = \frac12[\operatorname{tr} MM^\dagger - \frac1N|\operatorname{tr} M|^2] \le \frac{N}{2}$, and $|\nabla_\ell \operatorname{Re}\operatorname{tr} U_p|^2 \le |\nabla_\ell \operatorname{tr} U_p|^2 \le N/2$. Four links give $\sum_\ell |\nabla_\ell f|^2 \le 2N$ and

$$\Delta^{\mathrm{phys}}_{a,L} \;\le\; \frac{\hbar c\, g^2 N}{a\ \mathrm{Var}_0(\operatorname{Re}\operatorname{tr} U_p)}.$$

Corollary 3 (two-sided strong-coupling order). At $g = \infty$ the ground state is the Haar measure and $\mathrm{Var}(\operatorname{Re}\operatorname{tr} U_p) = \frac12\langle|\operatorname{tr} U|^2\rangle + \frac12\operatorname{Re}\langle(\operatorname{tr} U)^2\rangle$ equals 1 for $SU(2)$ and $\frac12$ for $SU(N \ge 3)$, by Peter–Weyl orthogonality of the fundamental character and the vanishing of $\int(\operatorname{tr} U)^2 dU$ for $N \ge 3$. The ground-state expectation of a local function is continuous in g^{-4} at $g = \infty$ uniformly in the volume (Yarotsky's Theorem 1, conclusion 4, analyticity of $\omega(A)$

in the perturbation parameter; the finite-volume statement is analytic perturbation theory of a simple isolated eigenvalue), so there is $g_1(N)$ with $\mathrm{Var}_0 \geq \frac{1}{4}$ for $g \geq g_1$. Combined with the strong-coupling note,

$$\gamma \frac{g^2}{2} \frac{N^2-1}{2N} \frac{\hbar c}{a} \;\leq\; \Delta_{a,L}^{\mathrm{phys}} \;\leq\; 4N\, g^2 \frac{\hbar c}{a} \qquad (g \geq \max(g_0, g_1)),$$

uniformly in the lattice size. The strong-coupling gap is $g^2\hbar c/a$ times a number bounded on both sides.

Why the plaquette says nothing about the continuum. As $a \to 0$ along the scaling curve the links approach the identity and $\mathrm{Var}(\mathrm{Re}\,\mathrm{tr}\, U_p) \to 0$ (the plaquette angle fluctuates on the scale g, and its square on the scale g^4), so the right side of Corollary 2 grows without bound. The plaquette is an ultraviolet observable whose excitation costs an energy of order $\hbar c/a$; the finiteness half $m < \infty$ of the conjecture requires an operator whose ratio $\sum_\ell \langle |\nabla_\ell O|^2 \rangle / \mathrm{Var}\, O$ stays of order a/ℓ for a physical length ℓ. The smeared-susceptibility treatment states the additional continuum hypotheses.

Abelian electric trials and the structure factor

For compact $U(1)$ write $U_\ell = e^{i\theta_\ell}$, $E_\ell = -i\partial/\partial\theta_\ell$, $\theta_p = \sum_{\ell \in \partial p} \epsilon_{p\ell}\theta_\ell$, and

$$H = \frac{\hbar c}{a}\Big[\frac{g^2}{2}\sum_\ell E_\ell^2 + \frac{1}{g^2}\sum_p (1 - \cos\theta_p)\Big]$$

(other conventions rescale g^2). Gauss's law generators $G_x = \sum_{\ell \ni x} \pm E_\ell$ commute with every E_ℓ, so

Lemma 4. $O_\varphi = \sum_\ell \varphi_\ell E_\ell$ with real φ is gauge invariant, and $\langle O_\varphi \rangle_0 = 0$.

Proof. The E_ℓ commute among themselves. Charge conjugation $\theta \mapsto -\theta$ is a unitary symmetry of H under which the unique ground state is invariant and $E_\ell \mapsto -E_\ell$. □

With $[E_\ell, h(\theta)] = -i\partial_\ell h$ and the lattice curl $(d\varphi)_p = \sum_{\ell \in \partial p} \epsilon_{p\ell}\varphi_\ell$: $[O_\varphi, \cos\theta_p] = i(d\varphi)_p \sin\theta_p$, $[O_\varphi, \sin\theta_p] = -i(d\varphi)_p \cos\theta_p$, hence

$$[O_\varphi, [H, O_\varphi]] = \frac{\hbar c}{a g^2}\sum_p (d\varphi)_p^2 \cos\theta_p,$$

and the variational identity above gives

$$\Delta^{\mathrm{phys}} \leq \frac{\hbar c}{2 a g^2}\frac{\sum_p (d\varphi)_p^2\, c_0}{\langle O_\varphi^2 \rangle_0}, \qquad c_0 = \langle \cos\theta_p \rangle_0,$$

where c_0 is independent of p by translation invariance of the unique ground state. To bound $\langle O_\varphi^2 \rangle$ below use the uncertainty relation with the gauge-invariant $\Theta = \sum_p \psi_p \sin\theta_p$: $\langle O^2 \rangle \langle \Theta^2 \rangle \geq \frac{1}{4}|\langle [O, \Theta] \rangle|^2$ with $\langle [O_\varphi, \Theta] \rangle = -ic_0 \sum_p \psi_p (d\varphi)_p$. Choosing $\psi = d\varphi$:

Theorem 5. If $c_0 > 0$, then for every real φ on the links with $d\varphi \neq 0$,

$$\Delta_{a,L}^{\mathrm{phys}} \;\leq\; \frac{2\hbar c}{a\, g^2\, c_0}\frac{\langle \Theta_\varphi^2 \rangle_0}{\sum_p (d\varphi)_p^2}, \qquad \Theta_\varphi = \sum_p (d\varphi)_p \sin\theta_p.$$

For a plane wave $\varphi_\ell = \cos(k \cdot x_\ell)\, u_{i(\ell)}$ the ratio is the structure factor $\hat{S}(k)$ of the plaquette sine at momentum k, and $\Delta^{\mathrm{phys}} \leq 2\hbar c\, \hat{S}(k)/(a g^2 c_0)$ for every $k \neq 0$ allowed by the box.

Proof. Insert the uncertainty bound into the previous display: $\langle O^2 \rangle \geq c_0^2 (\sum_p (d\varphi)_p^2)^2 / (4\langle \Theta^2 \rangle)$. □

Reading. In a Coulomb phase $\sin\theta_p$ is a curl of a nearly Gaussian field and $\hat{S}(k) \to 0$ as $k \to 0$; the theorem then forces $\Delta \to 0$ with the box, which is the gaplessness Guth and

Fröhlich–Spencer prove at weak coupling (abstract, B78), obtained here as a consequence of two ground-state quantities, c_0 and $\hat{S}$. In a confining phase $\hat{S}(0) > 0$ and the bound is finite, consistent with the strong-coupling gap. The theorem does not decide which phase occurs; it shows that vanishing of this structure factor is a sufficient criterion for gaplessness in the specified channel. A nonvanishing upper bound gives no converse lower-gap criterion.

Semisimple Gauss law and the limits of the dressing argument

Proposition 6. Let G have semisimple Lie algebra and $O = \sum_\ell \sum_a \varphi_\ell^a E_\ell^a$ with c-number coefficients. If O commutes with all Gauss generators, then $\varphi = 0$.

Proof. Write $E_\ell^a = L_\ell^a$ for the left generators of link ℓ, which satisfy $[L_\ell^b, L_\ell^a] = i f^{bac} L_\ell^c$, and let R_ℓ^b be the right generators, which commute with all L_ℓ^a. The Gauss generator at x is $G_x^b = \sum_{s(\ell)=x} L_\ell^b + \sum_{t(\ell)=x} R_\ell^b$ up to the sign convention. Then $[G_x^b, O] = i \sum_{s(\ell)=x} f^{bac} \varphi_\ell^a L_\ell^c$, with no contribution from the entering links. The L_ℓ^c are linearly independent operators, so $[G_x^b, O] = 0$ for all b, x forces $f^{bac}\varphi_\ell^a = 0$ for all b, c and every link, that is $[\varphi_\ell, T^b] = 0$ for all b: φ_ℓ lies in the centre of the Lie algebra, which is zero under the semisimplicity hypothesis. □

The proof identifies the exact boundary: coefficients must lie in the Lie algebra centre. A nonabelian group with a $U(1)$ factor still admits central abelian trials; the former unrestricted "nonabelian" statement was too broad.

Field-dependent gauge-covariant coefficients can produce invariant electric trials, with ordering/domain obligations. A Wilson-loop dressing is one example; local curvature coefficients are another. For such an operator, $[K, O]$ generally contributes to the double commutator, but O is a differential operator rather than a multiplication function. Proposition 1's gradient formula cannot be substituted for that extra term. The former universal positive cost proportional to $g^2|C|$ was not established. Gauge invariance removes the specified c-number semisimple trial; it proves neither a mandatory long string nor a nonvanishing dressing energy or lower gap.

Spatial Polyakov-loop trials

The averaged holonomy and its exact bound

Fix a direction i. The lattice decomposes into N_s^2 lines parallel to i, each of N_s links closing around the torus. For a line λ let $P_\lambda = \mathrm{Re}\,\mathrm{tr} \prod_{\ell\in\lambda} U_\ell$, the ordered product along the line; it is gauge invariant (a closed loop), and $\bar{P}_i = N_s^{-2} \sum_\lambda P_\lambda$. Let $O = \bar{P}_i - \langle \bar{P}_i \rangle_0$.

Polyakov-loop bound. For every $g > 0$, $N_s \geq 2$ and $SU(N)$ with the normalization of the obligations map,

$$\Delta_{a,L}^{\mathrm{phys}} \;\leq\; \frac{\hbar c g^2}{2a} \frac{\big\langle \sum_\ell |\nabla_\ell \bar{P}_i|^2 \big\rangle_0}{\mathrm{Var}_0\, \bar{P}_i} \;\leq\; \frac{\hbar c\, g^2 N}{4L\ \mathrm{Var}_0\, \bar{P}_i}.$$

Proof. O is a real gauge-invariant function of the links with zero mean, so Proposition 1 of the upper-bound note applies and gives the first inequality. Each link in direction i belongs to exactly one line and links in other directions do not enter $\bar{P}_i$, so $\nabla_\ell \bar{P}_i = N_s^{-2} \nabla_\ell P_{\lambda(\ell)}$. For a link of λ, $X_a \,\mathrm{tr} \prod U = \mathrm{tr}(T^a M)$ up to a phase with M a cyclic rearrangement of the product, and the completeness relation gives $\sum_a |\,\mathrm{tr}(T^a M)|^2 = \frac{1}{2}(N - |\,\mathrm{tr}\, M|^2/N) \leq N/2$; the real part has smaller gradient. Hence $\sum_\ell |\nabla_\ell \bar{P}_i|^2 \leq N_s^{-4} \cdot N_s^2 \cdot N_s \cdot N/2 = N/(2N_s)$, and $\frac{\hbar c g^2}{2a} \cdot \frac{N}{2N_s} = \frac{\hbar c g^2 N}{4L}$. □

Conditional small-volume comparison

For the formal weak-coupling constant-mode comparison, assume the links are near the identity and the gauge-invariant content of a Polyakov line is the holonomy of the constant mode: $\prod_{\ell\in\lambda} U_\ell \approx \exp(iLa_i)$ with $a_i = a_i^b T^b$ the constant mode of G07 Proposition 7, whose ground-state width is $|a_i| \sim g^{2/3}/L$ in natural units. Since $\mathrm{tr}\, T^b = 0$,

$$P_\lambda \approx \mathrm{Re}\,\mathrm{tr}\, e^{iLa_i} = N - \tfrac{1}{4} L^2 |\vec{a}_i|^2 + O(L^4|a|^4), \qquad \nabla_\ell P_\lambda \approx \mathrm{Im}\,\mathrm{tr}(T^a L a_i) = \tfrac{1}{2} L a_i^a,$$

so the variance of $\bar{P}_i$ is second order and the gradient first order in the constant mode:

$$\operatorname{Var}_0 \bar{P}_i \approx \tfrac{1}{16} L^4 \operatorname{Var}(|\vec{a}_i|^2), \qquad \Big\langle \sum_\ell |\nabla_\ell \bar{P}_i|^2 \Big\rangle_0 \approx N_s^{-4} \cdot N_s^3 \cdot \tfrac{1}{4} L^2 \langle |\vec{a}_i|^2 \rangle = \frac{L^2 \langle |\vec{a}_i|^2 \rangle}{4N_s}.$$

For the $SU(2)$ constant-mode model of C133, $La_i = g^{2/3}\xi_i$ with ξ_i distributed by the unit ground state $\psi_0^{(3)}$ of $\frac{1}{2}(-\Delta_{\mathbb{R}^9} + \sum_{i<j} |\vec{\xi}_i \times \vec{\xi}_j|^2)$:

$$\frac{\hbar c g^2}{2a} \cdot \frac{g^{4/3} \langle |\vec{\xi}_i|^2 \rangle / (4N_s)}{g^{8/3} \operatorname{Var}(|\vec{\xi}_i|^2)/16} = \frac{\hbar c}{L} g^{2/3} \cdot \frac{2 \langle |\vec{\xi}_i|^2 \rangle}{\operatorname{Var}(|\vec{\xi}_i|^2)}.$$

The bound is therefore $(\hbar c/L) g^{2/3}$ times the pure number $2\langle |\vec{\xi}_i|^2 \rangle / \operatorname{Var}(|\vec{\xi}_i|^2)$, finite because the unit ground state decays faster than any power along the valleys (the confining bound of C133 gives exponential decay of ψ_0 in $\sum_i |\vec{\xi}_i|$). Consistency with the exact zero-mode gap $\delta_1^{(3)} g^{2/3} \hbar c/L$ requires this number to be at least $\delta_1^{(3)}$, which is the Feynman–Bijl inequality applied inside the zero-mode model with the trial operator $|\vec{\xi}_i|^2$; the model version of the Polyakov-loop bound is exact and says exactly that.

What is proved and what is not: the Polyakov-loop bound is exact. The two approximations, replacing the ground-state expectations by their constant-mode values, are the leading order of the small-volume expansion (Lüscher; abstract, B78) and are not proved here. The statement that follows from them is that the exact lattice inequality carries the $g^{2/3}$ scaling of the small-volume corner on its upper side; a lower bound with that scaling remains the open weak-coupling target of STATE.

Infinite-volume susceptibility condition

Suppose the lattice theory at fixed a and g has infinite-volume gap $\Delta_a = \liminf_{N_s \to \infty} \Delta_{a,L} > 0$ (T2′), and write $mc^2 = \Delta_a$. the Polyakov-loop bound gives, for every L,

$$\operatorname{Var}_0 \bar{P}_i \;\le\; \frac{\hbar c g^2 N}{4L\, \Delta_{a,L}}.$$

By translation invariance in the transverse plane, $\operatorname{Var}_0 \bar{P}_i = N_s^{-2} \sum_y C_i(y)$ with $C_i(y) = \langle \operatorname{Re} P_\lambda(x) \operatorname{Re} P_\lambda(x+y) \rangle_c$ the connected two-point function of the spatial Polyakov loop at transverse separation y. Hence

Susceptibility consequence. If the infinite-volume gap is $mc^2 > 0$, then along any sequence of boxes on which $\Delta_{a,L} \ge mc^2/2$,

$$\sum_{y \in \text{transverse plane}} a^2\, C_i(y) \;\le\; \frac{\hbar c g^2 N}{2mc^2}\, L,$$

and the averaged spatial Polyakov loop self-averages at least like $1/L$.

Reading: in a confining phase the spatial Polyakov loop of length L has connected correlations decaying on a transverse scale ξ, and the left side is of order $\xi^2 \langle P^2 \rangle_c$, so the corollary is satisfied with room; in a phase where the loop orders or its correlations do not decay, the left side grows like L^2 and the corollary forbids a gap. It is a necessary condition only, but it is one written entirely in terms of a single gauge-invariant correlation and the coupling, with no reference to the spectrum.

The Haar strong-coupling comparison

At $g = \infty$ the links are independent Haar variables, distinct lines are independent, and $\langle |\operatorname{tr} U_1 \cdots U_{N_s}|^2 \rangle = 1$ for a product of independent Haar elements of $SU(N)$ (the product is

again Haar distributed), so $\operatorname{Var} P_\lambda = \frac{1}{2}$ for $N \geq 3$ and $\operatorname{Var} \bar{P}_i = 1/(2N_s^2)$. the Polyakov-loop bound then reads $\Delta \leq \hbar c g^2 N N_s^2/(2L)$, which grows with the lattice and says nothing. The averaged loop is the wrong trial operator in the disordered phase, as the plaquette is in the ordered one; the two bounds are complementary, and neither is uniform in a along the scaling curve.

Consequence for STATE

The exact upper bounds retain their specified trial states, groups, variances, units and finite-volume domains. Strong-coupling order, a vanishing abelian structure factor and a Polyakov susceptibility constraint answer different questions. Neither a nonzero upper bound nor the absence of one constant-colour trial establishes a lower gap. The small-volume spectral transport and the continuum equal-time smearing hypotheses remain open.